Chapter 7: Magnetostatics Fields

The following subjects will be discussed

- Magnetic Flux and Flux Density.

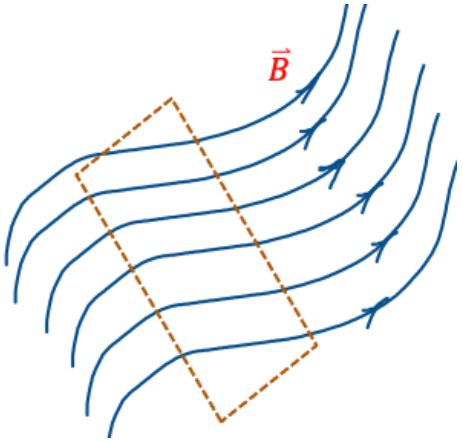
Magnetic Flux

Magnetic flux is defined as the number of magnetic field lines crossing a given surface.

Magnetic flux density is given by

$$\vec{B} = \mu_o \vec{H} \qquad (\mathrm{Wb/m^2 = T})$$

$\mu_o = 4\pi \mathrm{x} 10^{-7}$ (H/m) is the magnetic permeability of free space (vacuum).



Flux equals the surface integral of the field over the given surface.

(Wb) $$\psi = \int_S \vec{B} \cdot \overrightarrow{dS}$$

- $\vec{B}$: Magnetic flux density generated by current
- $\cdot$: Dot product
- $\overrightarrow{dS}$: Differential area
- $\int_S$: Surface integral

Note the "dot product": Only field components normal to the surface are counted!

For a closed surface $$\psi = \oint \vec{B} \cdot \overrightarrow{dA} = 0$$

Magnetic field lines go around closed paths, unlike electric fields!

Using divergence theorem, we get

$$\vec{\nabla} \cdot \vec{B} = 0$$

Magnetic Flux

Example: Find the magnetic flux from the wire 1 linking the given surface.

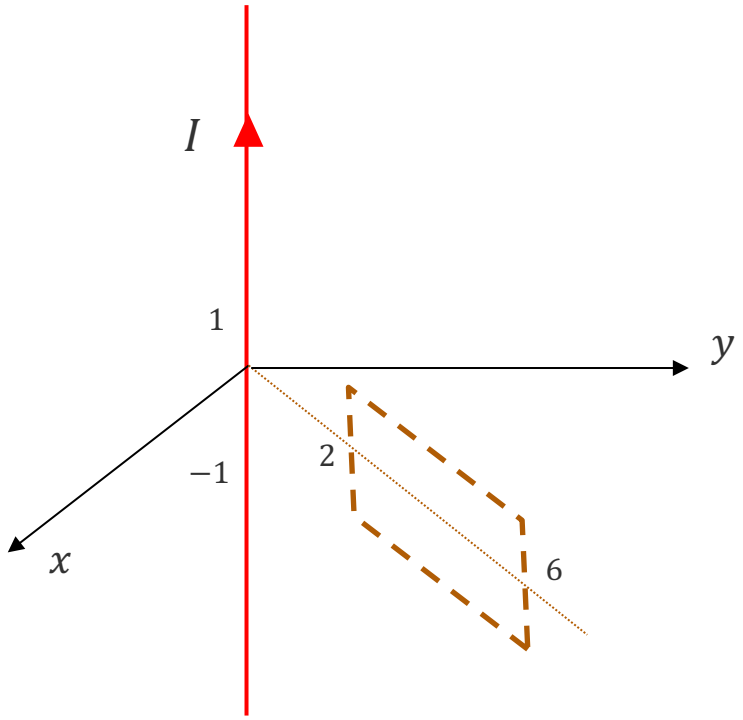
$$\vec{H} = \frac{I}{2\pi\rho}\vec{a}_\varphi \qquad \text{Infinite wire}$$

$$\vec{B} = \frac{\mu_o I}{2\pi\rho}\vec{a}_\varphi$$

$$\psi = \int_S \vec{B}\cdot\overrightarrow{dS} = \int_{-1}^{1}\int_{2}^{6} \frac{\mu_o I}{2\pi\rho}\, d\rho\, dz$$

$$\psi = \frac{\mu_o I\ (2)}{2\pi}\ln\left(\frac{6}{2}\right)$$

$$\psi = \frac{\mu_o I \ln 3}{\pi}$$



Chapter 8: Magnetostatics Forces

The following subjects will be discussed

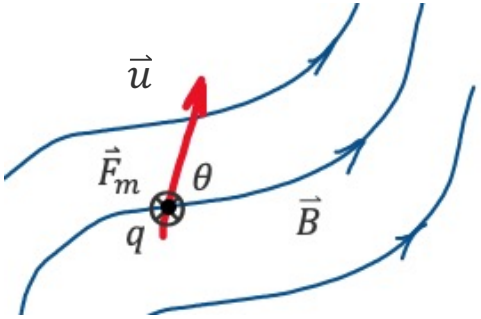
- Magnetic Flux and Flux Density Density.

Magnetic Forces: Isolated Charge

Consider an isolated charge q moving at constant velocity $\vec{u}$.

The electrostatic force is $\vec{F}_e = q\,\vec{E}$

The magnetostatic force is given by $\vec{F}_m = q\,(\vec{u} \times \vec{B})$



Use right-hand rule

- $\vec{F}_m$: Magnetic force (N)
- q: Charge + or – (C)
- $\vec{u}$: Charge velocity (m/s)
- $\times$: Cross product
- $\vec{B}$: Magnetic flux density (T)

Force magnitude is given by

$$|\vec{F}_m| = q\,|\vec{u}|\,|\vec{B}|\sin\theta$$

θ=0 ($\vec{u}$ and $\vec{B}$ in the same direction) → $\vec{F}_m$=0

The total force on the charge is given by $\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$

The trajectory of the charge can be found by solving the equations of motion.

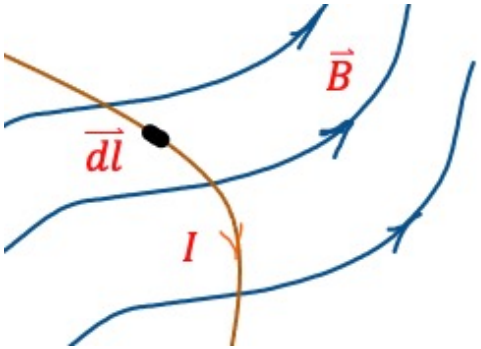
$$m\vec{a} = m\frac{d\vec{u}}{dt} = q(\vec{E} + \vec{u} \times \vec{B})$$

Magnetic Forces: Current Element

For current-carrying conductor, we have continuous moving charges (DC current).

Take a differential length and find moving charges.

$$I\,dl = \frac{dq}{dt}\,dl \quad = dq\,\frac{dl}{dt} \quad = dq\,u$$



Modify the basic force equation to a differential force

$$d\vec{F}_m = dq\,(\vec{u} \times \vec{B}) \;= I\,(\vec{dl} \times \vec{B})$$

Integrate over the current element (but only the part within the magnetic field).

Total magnetic force

$$\vec{F}_m = \int I\,(\vec{dl} \times \vec{B})$$

- Integration over conductor inside field
- Current in wire
- Differential length along direction of current
- Existing magnetic field density

Similarly

$$\vec{F}_m = \int (\vec{K}\,dS \times \vec{B})$$

$$\vec{F}_m = \int (\vec{J}\,dv \times \vec{B})$$

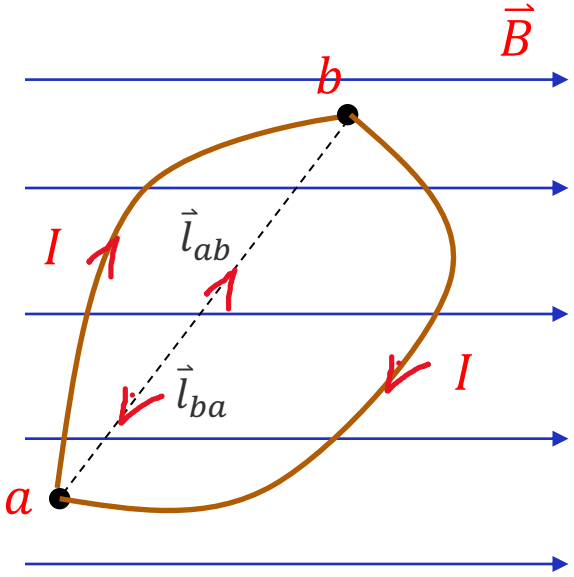
Magnetic Forces: Current Element

In case $\vec{B}$ is uniform and I is constant

$$\vec{F}_m = \int I\,(\overrightarrow{dl} \times \vec{B})$$

$$\vec{F}_m = I\left(\int_a^b \overrightarrow{dl}\right) \times \vec{B}$$

Vector Sum of length



$$\vec{F}_m = I\,(\vec{l}_{ab} \times \vec{B})$$

For a closed loop, $\vec{F}_m = 0$

Magnetic Forces: Current Element

Example. A current-carrying loop (half a circle, radius R) is placed in a region of magnetic field as shown. The current I and the magnetic flux density $\vec{B}$ are constants. Calculate force.

For the straight part,

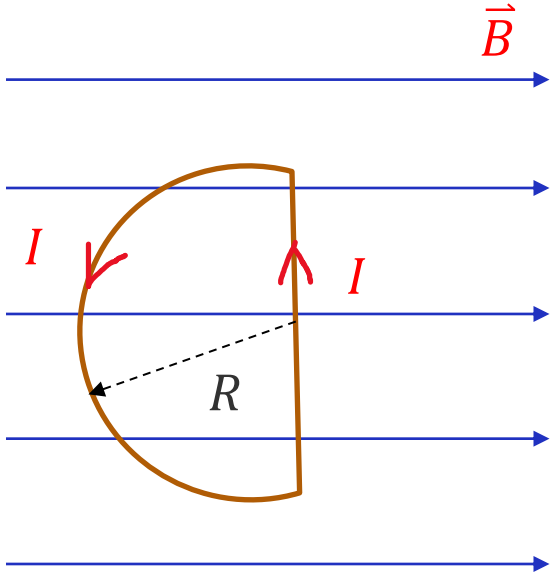
$$\vec{F}_m = I\left(\int_a^b \overrightarrow{dl}\right) \times \vec{B}$$

$$F_m = I\,(2R)B \qquad \text{Inside the page}$$

For the curved part,

$$F_m = I\,(2R)B \qquad \text{Outside the page}$$

$$\vec{F}_{m(tot)} = 0$$

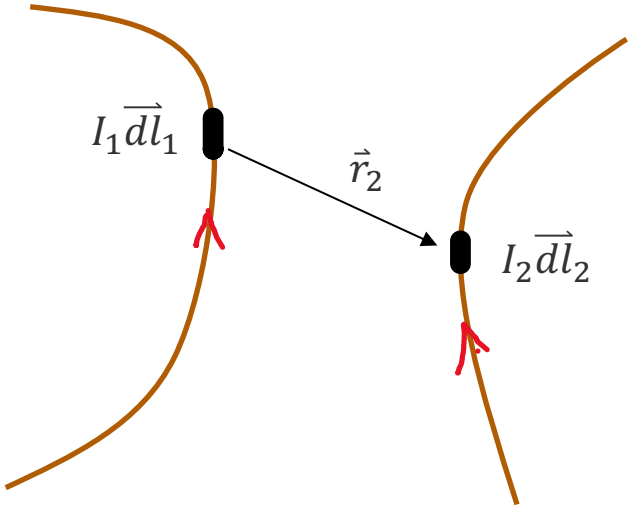


Magnetic Forces: Two Current Elements

Consider the force exserted by element 2 on element 1. In this case, the magnetic flux density in the force equation is that of element 2.

The differential field generated by the differential length of element 2 is given by

$$d\vec{B}_2 = \frac{\mu_o I_2}{4\pi}\frac{(\vec{dl}_2 \times \vec{a}_{r_2})}{r_2^2}$$



The differential force on element 1 is

$$d\vec{F}_1 = I_1(\vec{dl}_1 \times \vec{B}_2)$$

$$d(d\vec{F}_1) = I_1(\vec{dl}_1 \times d\vec{B}_2)$$

$$d(d\vec{F}_1) = \frac{\mu_o I_1 I_2}{4\pi}\frac{\vec{dl}_1 \times (\vec{dl}_2 \times \vec{a}_{r_2})}{r_2^2}$$

$$\vec{F}_1 = \frac{\mu_o I_1 I_2}{4\pi}\int_{l_1}\int_{l_2}\frac{\vec{dl}_1 \times (\vec{dl}_2 \times \vec{a}_{r_2})}{r_2^2}$$

$$\vec{F}_1 = -\vec{F}_2$$

Magnetic Forces: Two Current Elements

For two long (length l) parallel current-carrying wires, the magnetic force between them becomes,

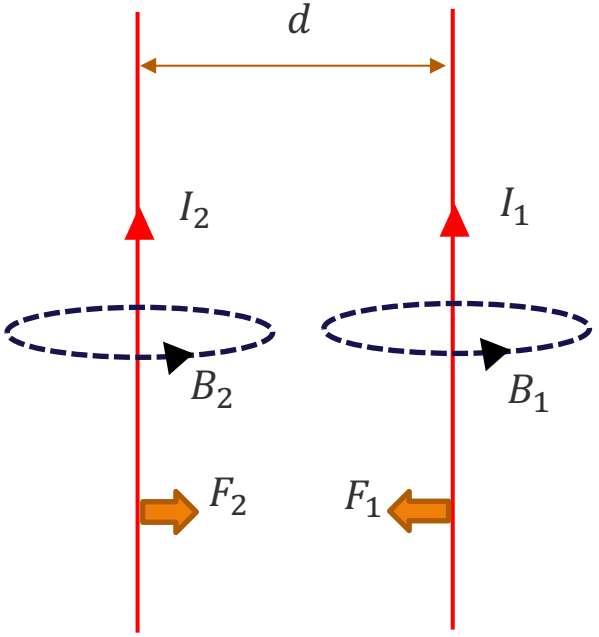
$$d\vec{F}_1 = I_1(\vec{dl}_1 \times \vec{B}_2)$$

$$d\vec{F}_1 = I_1\left(dz\,\vec{a}_z \times \frac{\mu_o I_2}{2\pi d}\vec{a}_\varphi\right)$$

$$\vec{F}_1 = -\vec{a}_\rho \frac{\mu_o I_1 I_2}{2\pi d}\int_0^l dz$$

$$\vec{F}_1 = \frac{\mu_o I_1 I_2 l}{2\pi d}(-\vec{a}_\rho)$$

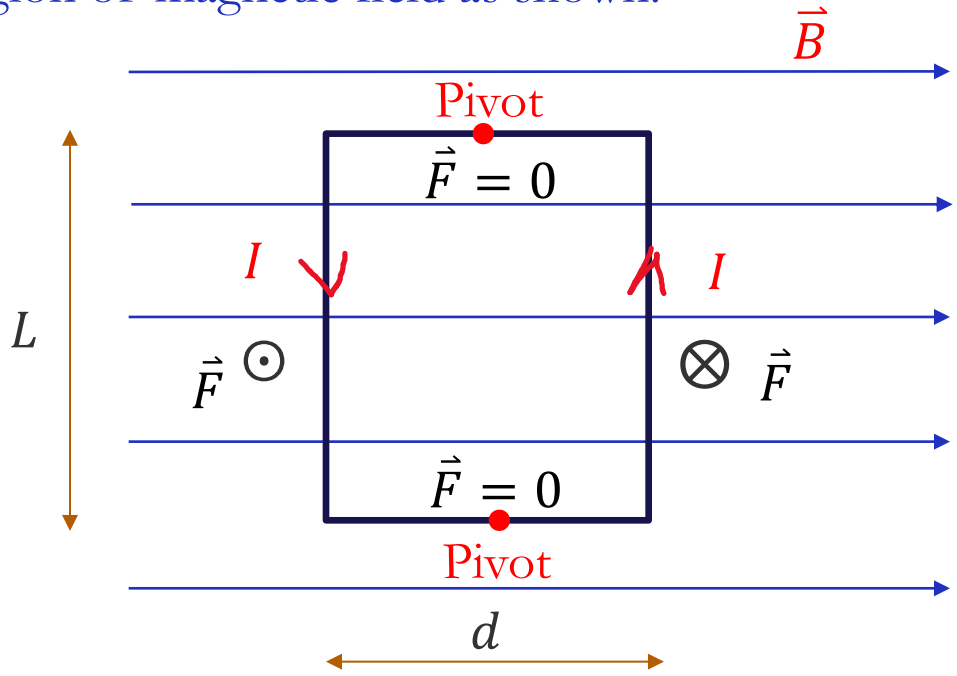
$$\vec{F}_2 = \frac{\mu_o I_1 I_2 l}{2\pi d}(\vec{a}_\rho)$$ Attractive force

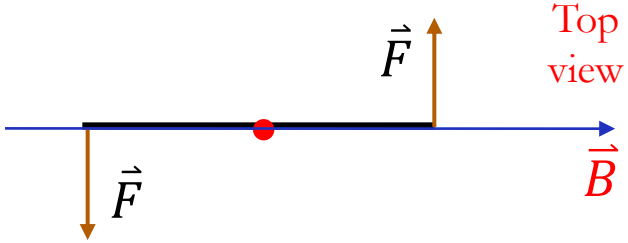


If the currents are opposite, the force is repulsive.

Magnetic Forces: Current Element

Application. Consider a rectangular loop carrying constant current and placed in a region of magnetic field as shown.

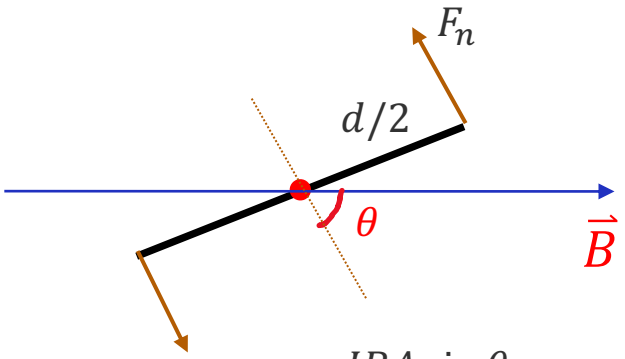




$$F = I\,L\,B$$

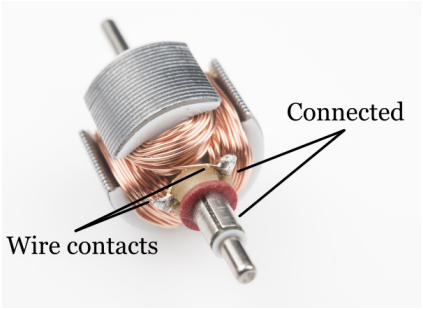
$$\tau = 2\,I\,L\,B\,(d/2)$$

$$\tau = IBA$$



$$\tau = IBA\sin\theta$$

The concept of electric motors!





Magnetic Force on a Current-carrying Conductor

Example. Consider a rectangular coil with 25 turns, carrying constant current of 15 mA with an area of (5.4x8.5) cm^2. The coil is placed in a region of constant magnetic field of 0.35 T. The coil area is perpendicular to the field direction.

Calculate the torque on the coil.

$$\tau = INBA\sin\theta$$

$$\tau = (15\times10^{-3})(25)(0.35)(5.4\times8.5)\times10^{-4}(\sin 90°)$$

$$\tau = 6.02\times10^{-4}\ \text{N.m}$$

Magnetic Force Between Wires

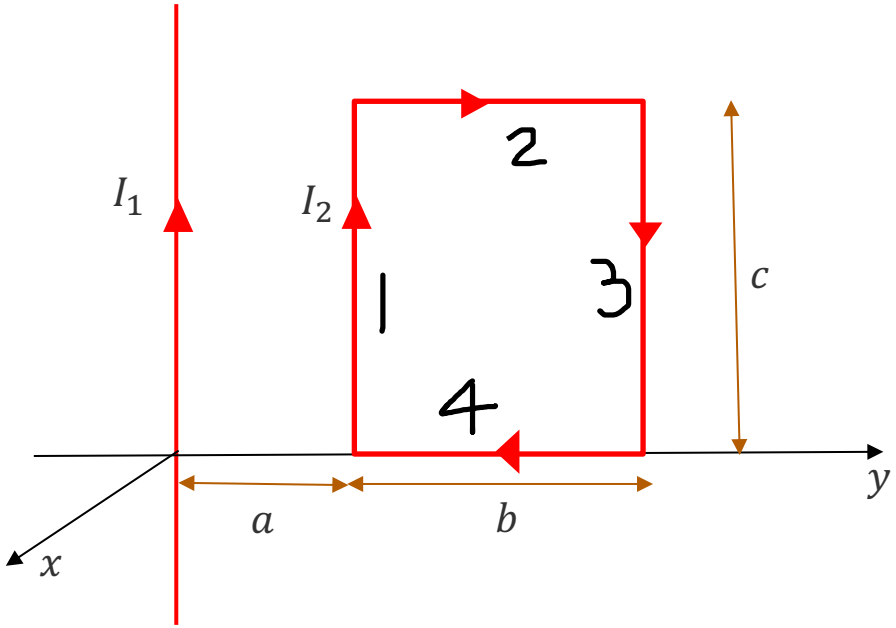
Example: Find the total force on wire 2 due to wire 1. Wire 1 is very long.

$$\vec{B}_1 = \frac{\mu_o I_1}{2\pi y}(-\vec{a}_x)$$ Infinite wire

Force equation $$\vec{F}_2 = \int I_2 (\vec{dl}_2 \times \vec{B}_1)$$

B_1 is uniform for s_1 and s_3.

B_1 is not uniform for s_2 and s_4.



$$\vec{F}_{s1} = \frac{\mu_o I_1 I_2}{2\pi a}\int_0^c dz\, \vec{a}_z \times (-\vec{a}_x) = \frac{\mu_o I_1 I_2}{2\pi}\frac{c}{a}(-\vec{a}_y)$$

$$\vec{F}_{s3} = \frac{\mu_o I_1 I_2}{2\pi(a+b)}\int_0^c dz\, (-\vec{a}_z) \times (-\vec{a}_x) = \frac{\mu_o I_1 I_2}{2\pi}\frac{c}{(a+b)}(\vec{a}_y)$$

$$\vec{F}_{s2} = I_2 \int_a^{a+b} \frac{\mu_o I_1}{2\pi y}\, dy\, \vec{a}_y \times (-\vec{a}_x) = \frac{\mu_o I_1 I_2}{2\pi} \ln\left(\frac{a+b}{a}\right)(\vec{a}_z)$$

$$\vec{F}_{s4} = -\vec{F}_{s2}$$

$$\vec{F}_{2_net} = \vec{F}_{s1} + \vec{F}_{s2} + \vec{F}_{s3} + \vec{F}_{s4}$$

Loop 2 moves left!

How to make it move right?

Q & A

Send me an email if you have any questions.